How Tariffs Reshape Supply Chains: Canada's 2024 Surtax on Chinese Steel

Xiang Zheng

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Summary

After Canada's 25% surtax on Chinese steel took effect in October 2024, China's share of Canada's steel imports fell from about 6% to under 4%. A difference-in-differences comparison of steel products on and off the surtax list suggests that the decline occurred mainly in listed products. The estimated effect is a decline of about 2 percentage points, ranging from about 1 to 3 points depending on the model. The estimates are not strong evidence on their own:

- The cleanest estimate uses only the four months before the 2025 U.S.-Canada tariff dispute. It is about -1.9 points and not statistically significant at the 5% level ($p = 0.086$).
- The more precise estimate (about -3.1 points) uses a longer window that overlaps with the tariff dispute.
- In the China-only time series, adding a linear trend reduces the estimate to about -1.1 points, which is not statistically significant.

1. Question and background

In the offshoring model, a firm decides where to source each input by comparing the cost of producing it at home with the cost of buying it abroad, including trade costs. A tariff raises the trade cost of buying from the targeted country, so the model predicts that buyers shift to other suppliers. When there are several foreign suppliers, part of this shift goes to third countries (trade diversion).

On August 26, 2024, Canada announced a 25% surtax on steel and aluminum products from China, and the surtax took effect on October 22, 2024. This analysis asks:

1. Did China's share of Canada's steel imports fall after the surtax?
2. If so, which countries' shares rose instead?
3. Can the change be attributed to the surtax, or to other trends and policies?

Several other steel policies followed, so they have to be separated from the surtax:

Date	Event
Aug 26, 2024	25% surtax on Chinese steel and aluminum announced
Oct 22, 2024	Surtax takes effect

Date	Event
Mar 12-13, 2025	U.S. imposes 25% tariffs on Canadian steel; Canada imposes 25% counter-tariffs on U.S. steel
Jun 27, 2025	Tariff-rate quotas on steel from countries without a free trade agreement with Canada (including China and Taiwan; Japan, South Korea and Vietnam are excluded)
Jul 31, 2025	25% surtax on goods containing steel melted and poured in China
Aug 1, 2025	Quotas tightened for non-FTA countries and extended to FTA partners other than the U.S. and Mexico

2. Data

Monthly import values (CAD) and quantities for HS chapter 72 (iron and steel) at the 6-digit product level, January 2023 to August 2026, from Statistics Canada's Canadian International Merchandise Trade (CIMT) Web Application. Files were downloaded separately for the world total and for six source countries: China, the United States, South Korea, Japan, Vietnam and Taiwan. Quantities reported in kilograms were converted to tonnes.

3. Import shares

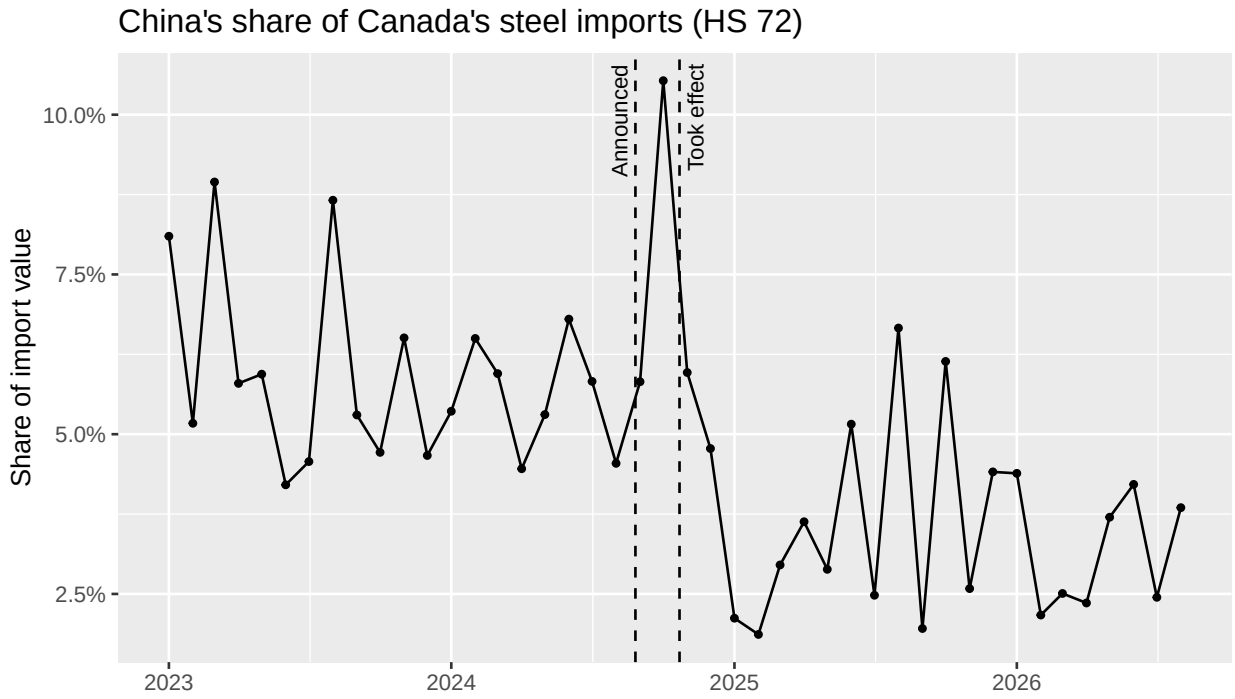
Each country's share is its monthly import value (or tonnes) divided by Canada's total monthly steel imports from the world.

As a check, the six countries together account for 60% to 82% of Canada's steel imports in every month, so the shares are below 100% as they should be.

Table 1: Combined import share of the six source countries

Minimum	Average	Maximum
60.4%	68.8%	82.0%

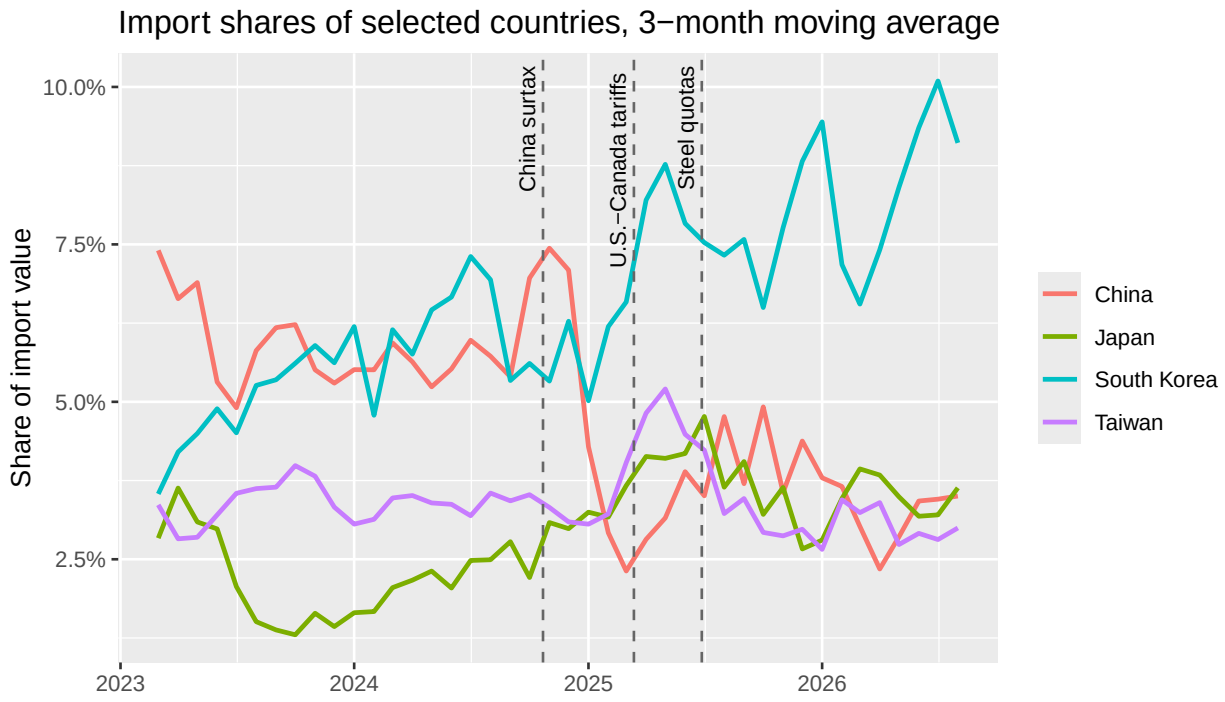
4. China's share over time



China's share fluctuated between roughly 4% and 9% before the surtax, drifting down during 2023 and staying around 5% to 6% through most of 2024. It peaked at about 10.5% in September 2024, between the announcement and the effective date. Section 11 shows a similar spike for products not covered by the surtax, so the peak may not reflect stockpiling ahead of the surtax. After the surtax took effect, the share fell to around 2% in early 2025 and has mostly stayed between 2% and 4.5% since.

5. Which countries' shares rose?

To reduce month-to-month noise, the chart below uses a three-month moving average. The U.S. is shown separately because its share (around 50%) is on a different scale.



Japan's and South Korea's shares were already rising before the surtax: Japan's from about 1.3% in late 2023 to about 3% by October 2024, and South Korea's from about 3.5% in early 2023 to about 6% to 7% in 2024. Their increases after the surtax therefore cannot be attributed to trade diversion alone.

6. Separating the surtax from later policies

The period after the surtax contains several policy changes, so it is split into five stages:

1. Before: January 2023 to July 2024
2. Transition: August to October 2024, between the announcement and the effective date
3. Surtax only: November 2024 to February 2025
4. US tariffs: March to July 2025, during the U.S.-Canada tariff dispute
5. Quotas: August 2025 to August 2026, with tariff-rate quotas in place

Table 2: Average monthly share of Canada's steel import value

country	1. Before	2. Transition	3. Surtax only	4. US tariffs	5. Quotas
China	5.9%	7.0%	3.7%	3.4%	3.6%
Japan	2.2%	2.2%	3.6%	4.4%	3.4%
South Korea	5.4%	5.6%	5.7%	7.9%	8.2%
Taiwan	3.4%	3.5%	3.2%	4.7%	2.9%
United States	52.0%	48.5%	53.2%	44.6%	49.9%
Vietnam	0.7%	0.3%	0.5%	0.7%	0.7%

- **China's share fell as soon as the surtax took effect** (from 5.9% to 3.7%) and stayed around 3.4% to 3.6% afterwards. The later policies did not push it much lower.

- **Japan's share rose from 2.2% to 3.6%** in the surtax-only period, roughly matching China's decline. Because Japan's share was already rising before the surtax, only part of this increase is likely to reflect diversion away from China.
- **The U.S. decline came later.** The U.S. share was unchanged in the surtax-only period (53.2%) and fell only during the 2025 tariff dispute (44.6%), before recovering.
- **South Korea's and Taiwan's shares moved with the 2025 policies.** South Korea's share was flat in the surtax-only period (5.7%) and rose to about 8% from March 2025. Taiwan's share fell from 4.7% to 2.9% once quotas applied to countries without a free trade agreement, a group that excluded South Korea and Japan.
- **Vietnam's share stayed small and stable**, so there is no clear sign of Chinese steel being rerouted through Vietnam at this level of detail.

7. Seasonality check

The surtax-only period covers winter months, so it is compared with the same months one year earlier.

Table 3: Same winter months, one year apart (share of value)

country	Nov 2023 - Feb 2024 (no surtax)	Nov 2024 - Feb 2025 (surtax)
China	5.8%	3.7%
Japan	1.7%	3.6%
South Korea	5.6%	5.7%
United States	52.6%	53.2%

In the same winter months, China's share fell from 5.8% to 3.7%, while South Korea and the U.S. were essentially unchanged. The decline is not a seasonal pattern. Japan's share rose from 1.7% to 3.6%, but part of this continues its earlier upward trend.

8. Robustness: tonnes instead of value

Steel prices moved a lot over this period, so value shares could partly reflect price changes. The same comparison using tonnes:

Table 4: Same winter months, share of tonnes

country	Nov 2023 - Feb 2024 (no surtax)	Nov 2024 - Feb 2025 (surtax)
China	5.1%	3.2%
Japan	0.7%	1.9%
South Korea	6.0%	6.3%
United States	51.4%	51.0%

The pattern holds in tonnes: China fell from 5.1% to 3.2% and Japan rose from 0.7% to 1.9%. Japan's value share is much higher than its tonnage share, which suggests that imports from Japan are higher-priced per tonne, possibly higher-value products.

9. Time-series regression

A linear regression of China's monthly value share on three policy indicators, with calendar-month fixed effects to control for seasonality. The indicators are cumulative, so `us_tariff` and `quota` measure changes in addition to the surtax. The August-October 2024 months are excluded.

Table 5: Policy coefficients (calendar-month fixed effects not shown; n = 41, R-squared = 0.617)

term	estimate	std_err	lower	upper	p_value
surtax	-0.0241	0.0086	-0.0418	-0.0064	0.0096
us_tariff	0.0010	0.0111	-0.0217	0.0238	0.9253
quota	-0.0027	0.0080	-0.0192	0.0138	0.7415

Controlling for seasonality and the later policies, the surtax is associated with a decline of about 2.4 percentage points in China's share (p = 0.0096; 95% CI -4.2 to -0.6 percentage points). For comparison, China's average share before the surtax was 5.9% (Table 2). The U.S. tariff and quota indicators are small and not statistically significant.

Adding a time trend

China's share was already drifting down in 2023, so the regression is re-estimated with a linear time trend.

Table 6: Surtax coefficient with and without a time trend

specification	estimate	lower	upper	p_value
Without trend	-0.0241	-0.0418	-0.0064	0.0096
With linear trend	-0.0110	-0.0359	0.0139	0.3724

With a linear trend, the estimate falls to about -1.1 percentage points and is no longer statistically significant (p = 0.37). With a short series in which all post-surtax months come late in the sample, the trend and the policy indicators are hard to separate. A linear trend may also over-control, since China's share was roughly flat through 2024 and then dropped sharply. This motivates the difference-in-differences design in Section 11, which uses products not on the surtax list as a comparison group.

Bootstrap confidence interval

Table 7: 95% confidence intervals for the surtax coefficient (no trend)

method	lower	estimate	upper
Normal theory (confint)	-0.0418	-0.0241	-0.0064
Bootstrap percentile	-0.0494	-0.0241	0.0011

The bootstrap interval (-4.9 to +0.1 percentage points) is wider than the normal-theory interval and just includes zero, reflecting how few surtax-only months the estimate rests on.

10. Ratio estimator vs. mean of monthly shares

The mean of monthly shares weights every month equally, while the ratio estimator (total imports from China divided by total imports from the world over the period) weights months by their import value.

Table 8: China's share: two ways to estimate

window	mean_of_shares	ratio_estimator
Nov 2023 - Feb 2024	5.8%	5.8%
Nov 2024 - Feb 2025	3.7%	3.7%

Both methods give the same result (5.8% and 3.7%), so the finding does not depend on how the shares are averaged.

11. Difference-in-differences: products on vs. off the surtax list

The surtax applies only to specific tariff items. The Department of Finance's final list (October 2024) includes 130 tariff items in chapter 72. All of them are 8-digit items ending in ".00", which in Canada's Customs Tariff means the item covers its whole 6-digit HS subheading. Each listed item therefore matches exactly one 6-digit code in the Statistics Canada data, with no partial coverage.

The 37 chapter-72 codes in the data that are not on the list are:

- 30 codes in headings 7201 to 7205: pig iron, ferro-alloys, direct-reduced iron, scrap, and granules and powders.
- 7 codes in headings 7206 to 7229: 7214.10 (forged bars), 7216.61, 7216.69 and 7216.91 (certain angles and shapes), 7220.11 (narrow hot-rolled stainless flat products), 7228.10 (high-speed steel bars) and 7228.80 (hollow drill bars).

If the surtax drove the decline, China's share should fall for listed products but not for unlisted ones. Month fixed effects absorb anything that affects both groups in the same month, including overall trends, steel prices and Canadian demand.

Table 9: Product groups before the surtax (Jan 2023 - Jul 2024)

group	products	world_value_bn	china_value_bn	china_share
Not on list	37	1.630	0.077	0.047
On surtax list	130	18.415	1.113	0.060

Table 10: Chapter 72 codes in headings 7206-7229 that are not on the surtax list

hs6	description
7214.10	Bars & rods, iron or non-alloy steel forged
7216.61	Angles, shapes & sect,i/nas,nfw than cf or cfi, obtained ...
7216.69	Angles, shapes and sections, i/nas, nfw than cold formed ...
7216.91	Angles, shapes and sections, i/nas, cf or cold finished f...
7220.11	Flat rolled products,stainless steel,hot rolled, < 600 mm...
7228.10	Bars and rods of high speed steel, nes
7228.80	Bars & rods, hollow drill, alloy or non-alloy steel

Before the surtax, China supplied 6.0% of listed products and 4.7% of unlisted products. The unlisted group is much smaller (about C$1.6 billion over 19 months, compared with C$18.4 billion), so its monthly share is noisier.

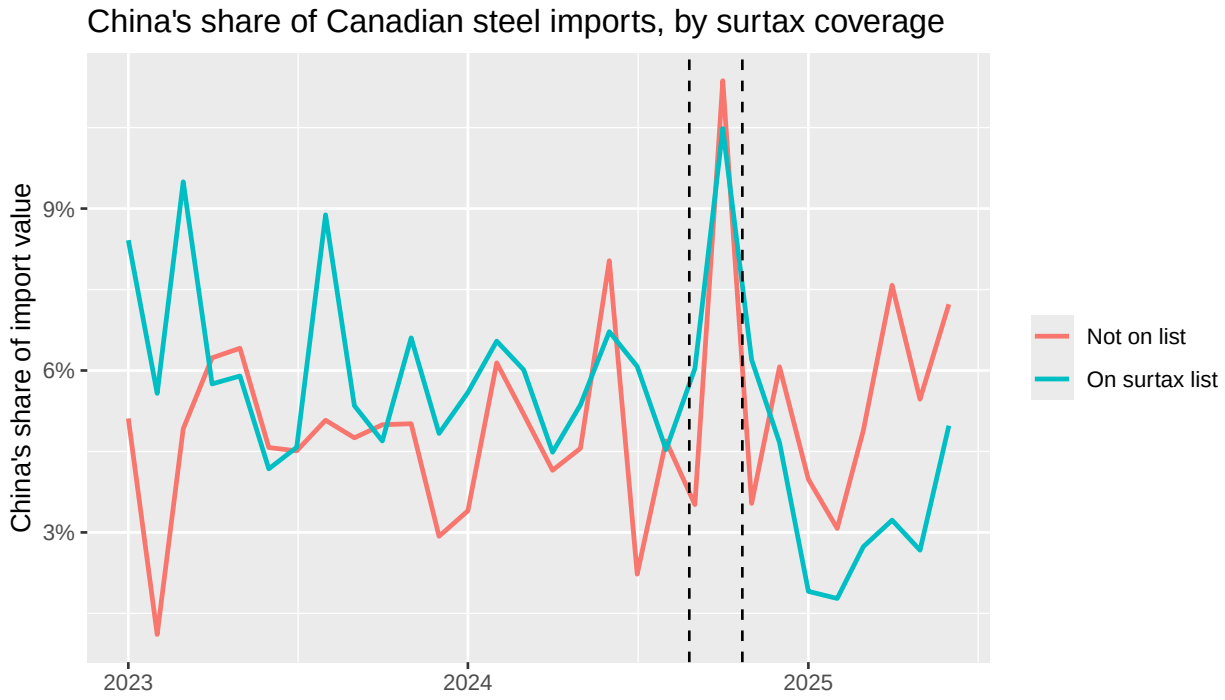


Table 11: Difference-in-differences estimate of the surtax effect

window	estimate	lower	upper	p_value
Jan 2023 - Jan 2025	-0.0163	-0.0413	0.0088	0.1911
Jan 2023 - Feb 2025 (main)	-0.0188	-0.0405	0.0029	0.0858
Jan 2023 - Jun 2025	-0.0306	-0.0471	-0.0142	0.0008

Before August 2024, the two series move in roughly parallel paths between about 4% and 7%, with the smaller unlisted group noisier. Both spike in September 2024, so that spike is not specific to the

products targeted by the surtax. After the surtax took effect, China's share of listed products fell to about 2% to 3% and stayed there, while its share of unlisted products returned to around 4% to 7%.

The difference-in-differences estimate is about -1.9 percentage points in the main window, which ends before the U.S.-Canada tariff dispute (95% CI -4.1 to +0.3; $p = 0.086$). In the longer window through June 2025, the estimate is about -3.1 percentage points (95% CI -4.7 to -1.4; $p < 0.001$). The longer window is more precise but also includes Canada's March 2025 counter-tariffs on U.S. steel, which mainly affected steel products and could have influenced the listed group.

February 2025 may also be affected by the dispute. The U.S. announced tariffs on Canadian goods on February 1, 2025 and paused them two days later, and on February 10 it announced 25% tariffs on steel imports starting in March. Ending the window in January 2025 leaves three post-surtax months and gives an estimate of -1.6 percentage points (95% CI -4.1 to 0.9; $p = 0.19$).

12. Limitations

- **Short post-surtax window.** Only four months separate the surtax from the U.S.-Canada tariff dispute, so the cleanest estimates are imprecise. February 2025 may already reflect expectations of U.S. tariffs.
- **Sensitivity to trends.** In the China-only time series, the surtax estimate shrinks and loses significance once a linear trend is added. The difference-in-differences design addresses this by comparing product groups within the same months, but relies on the two groups following parallel trends.
- **Comparison group.** Most unlisted products are raw materials (pig iron, ferro-alloys, scrap), which serve different markets from finished steel. The group is also much smaller, so its monthly share is noisier.
- **Inference.** With two groups and monthly data, standard errors may be understated because of autocorrelation. Newey-West standard errors or a block bootstrap would be more appropriate.
- **Coverage.** The surtax also applies to some products outside chapter 72 (for example, pipes and tubes in chapter 73), which are not analyzed here.
- **Country of origin.** Customs data record country of origin. If Chinese steel was processed or rerouted through third countries, it may be recorded under another origin. Canada's July 2025 rule on steel melted and poured in China suggests this was a policy concern.
- **Census, not sample.** Customs data are a complete record rather than a survey, so the main uncertainty comes from month-to-month variation, not sampling error.

13. Conclusion

The evidence is consistent with the offshoring model's prediction that raising the trade cost of a supplier shifts sourcing away from it. China's share of Canada's steel imports fell sharply when the surtax took effect, and the decline was not seasonal or driven by prices. It appeared mainly in the products the surtax targeted. The estimated effect is a decline of about 2 percentage points in China's share of listed products, between about 1 and 3 points depending on the model, and the cleanest estimate is not statistically significant at the 5% level. Identifying where the lost share went is harder: Japan's and South Korea's shares rose, but both were already rising before the surtax, and later shifts in U.S., South Korean and Taiwanese shares line up with the 2025 tariff dispute and import quotas. Separating the surtax from these overlapping trends and policies is the main challenge, and the results should be read with that uncertainty in mind.

Data sources

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